Extension: Perpendiculars and distances

1 Find the perpendicular distance d from the point (x_1, y_1) to the line $Ax + By + C = 0$.

2 Find the perpendicular distance from the origin to the line $4x + 7y + 5 = 0$.

3 Consider two parallel lines $9x - 9y - 9 = 0$ and $9x - 9y - 0 = 0$:

- **a** Find the y-coordinate of the point on the line $9x - 9y - 9 = 0$ whose x-coordinate is 8.
- **b** Use the point obtained from part(a) to find the shortest distance between the two given lines.

4 If the perpendicular distance from the point $(-5, -1)$ to the line $-2x - 7y + k = 0$ is $\dfrac{10}{\sqrt{53}}$, find the value(s) of k.

5 If the perpendicular distance from the point $(6, k)$ to the line $6x + (-2y) + (-6) = 0$ is $\dfrac{40}{\sqrt{40}}$, find the value(s) of k.

6 Consider the given point and the corresponding line:
Find the distance from the point to the line.

- **a** $(3, 5)$, $4x + 3y + 1 = 0$
- **b** $(4, -3)$, $24x + 7y - 3 = 0$
- **c** $(5, 4)$, $5x - 12y = 0$
- **d** $(-5, 3)$, $3x - 2y - 5 = 0$

7 Consider the points $A\,(5, 7)\,, B\,(9, 10)\,, C\,(6, 4)$ and $D\,(2, 1)$, which are the vertices of a parallelogram.

- **a** Determine the equation of the line going through AB. Give your answer in the form $ax + by + c = 0$.
- **b** If AB is the length of the parallelogram, find the perpendicular height of the parallelogram.
- **c** Hence, find the exact area of the parallelogram.

8 Consider the lines L_1: $x + 2y + 5 = 0$ and L_2: $x + by + 20 = 0$.

- **a** Find the distance between L_1 and the origin.
- **b** Solve for the value(s) of b such that L_1 is twice as far from the origin as L_2.